REVIEW 1: PROJECT SETUP, SRS & WORKING APPLICATION DEMO

AegisPharm Enterprise

Smart Clinical Pharmacy Management System with Total Quality Management (TQM) Architecture

PROJECT PRESENTER:

👤 **Piyush Bhatt**

Student, Amrapali University, Haldwani, Uttarakhand

EVALUATION MILESTONE:

🏛️ **Amrapali University**

Evaluation: Review 1 — Setup, SRS & Working Application

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RUBRIC: PROBLEM STATEMENT & BASELINE

Problem Statement & Baseline Clinical Understanding

🚨 1. MEDICATION DISPENSING ERRORS

• Clinical Reality: Over 1.3 million preventable patient injuries occur annually due to adverse drug events (ADEs).

• Handwriting & Transcription: Manual interpretation of handwritten prescriptions causes dosage miscalculations.

• Allergy Blindspots: Retail pharmacies lack real-time EHR cross-checks, risking fatal Penicillin or NSAID anaphylaxis.

• Baseline Remedy: Automated Clinical Decision Support System (CDSS) acting as an immutable quality shield.

❄️ 2. COLD-CHAIN & EXPIRY LOSS

• Biological Potency Loss: Vaccines, insulin, and biologics degrade silently outside the 2.0°C–8.0°C thermal envelope.

• FIFO Picking Flaw: Standard retail FIFO allows older batches to sit unnoticed while newer stock is sold.

• Dead-Stock Write-Offs: Significant annual capital lost in expired, unsellable inventory.

• Baseline Remedy: Algorithmic FEFO (First-Expiry, First-Out) picking + 24/7 continuous IoT temperature telemetry.

⚖️ 3. NARCOTIC COMPLIANCE & TILL DRIFT

• Schedule X / NDPS Violations: Inadequate tracking of habit-forming sedatives invites severe legal penalties and license revocation.

• Cash Drawer Shrinkage: Till variances, untracked mid-shift cash drops, and unverified digital UPI receipts.

• Failed Inspector Audits: Inability to generate instant batch reports during surprise CDSCO/FDA inspections.

• Baseline Remedy: Mandatory dual-PIN narcotic register + Daily Z-Report mathematical drawer reconciliation.

RUBRIC: QUALITY OBJECTIVE

Quality Objectives Aligned with TQM Principles

Objective 1: Zero Medication Dispensing Defects (Patient Focus)

Poka-Yoke error-proofing: Implement an automated Clinical Decision Support System (CDSS) that intercepts allergen cross-reactions (Penicillin, NSAIDs, Sulfa) and toxic drug-drug interactions before an invoice can be created.

Objective 2: 100% Cold-Chain Storage & Transport Compliance

Biological Integrity: Guarantee all thermo-sensitive pharmaceuticals are stored within the mandatory 2.0°C–8.0°C range via real-time 30s IoT telemetry streaming and automatic thermal spike alarms.

Objective 3: Zero Expiry Dead-Stock Losses (FEFO Scheduling)

Process Quality: Eliminate inventory wastage by algorithmically enforcing First-Expiry, First-Out (FEFO) batch prioritization, ensuring nearest-expiry viable medicines are dispensed first.

Objective 4: Zero Cash Register Discrepancy & Till Balancing

Fact-Based Management: Enforce strict shift-end financial governance through the Daily Z-Report engine, mathematically balancing opening float, sales, payouts, and counted physical currency.

Objective 5: 100% Regulatory Audit Readiness (ISO 9001 & CDSCO)

Continuous Assurance: Maintain an immutable SQLite audit trail, dual-PIN Schedule X narcotic register, and 1-click CSV export ready for State Drug Inspectors.

RUBRIC: SRS (REQUIREMENTS ENGINEERING)

Software Requirements Specification (SRS) — IEEE 830

📋 FUNCTIONAL REQUIREMENTS (FR)

• FR-01 (Authentication & RBAC): System shall enforce role-locked workstation sessions across 5 distinct staff personas.

• FR-02 (eRx Intake & CDSS): System shall validate incoming digital prescriptions and trigger blocking alerts on allergen contraindications.

• FR-03 (Schedule X Narcotics): System shall require Doctor DMC Reg No. and 4-digit Pharmacist PIN verification for controlled substances.

• FR-04 (POS & Multi-Tender): System shall support 1-click verified eRx import, barcode search, dynamic UPI QR generation, and cash change calculation.

• FR-05 (FEFO Batch Picker): System shall sort inventory batches by earliest calendar expiry and prevent dispensing expired SKUs.

• FR-06 (Cold-Chain IoT): System shall log digital temperature every 30s and trigger visual/audio alarms when temperatures exceed 8.0°C.

• FR-07 (Delivery e-POD): System shall require a 4-digit customer SMS OTP before an order can be marked as 'Delivered'.

• FR-08 (Z-Report Close): System shall reconcile shift cash, UPI, card, and CGST/SGST tax liabilities into signed, immutable records.

⚡ NON-FUNCTIONAL REQUIREMENTS (NFR)

• NFR-01 (Performance & Latency): System shall execute database queries in <50ms and render desktop UI at 60 FPS via Edge Chromium WebView2.

• NFR-02 (Reliability & ACID Compliance): SQLite database shall ensure 100% ACID compliance with automatic timestamped snapshot backups.

• NFR-03 (Security & Tamper-Proofing): System shall record all user actions in an immutable append-only audit trail and prevent unauthorized role switching.

• NFR-04 (Maintainability & Modularity): System shall strictly follow a 6-tier modular folder architecture with decoupled bridge RPC layers.

• NFR-05 (Regulatory Compliance): System shall comply with Indian Drugs & Cosmetics Act (Schedule H1/X) and GST electronic invoicing standards.

• NFR-06 (Portability): System shall run natively as a high-performance Windows desktop application and as a modern web application.

RUBRIC: SCOPE & USER ROLES

Project Scope & Granular User Roles (RBAC)

👑 Store Admin

Terminal: Console HQ

- Manage staff accounts & passwords
- Monitor live online terminal presence
- Review daily Z-Report reconciliations
- Execute SQLite snapshot backups
- Inspect system audit stream

🩺 Clinical Pharmacist

Terminal: Clinical Bay

- Review incoming e-Prescriptions queue
- CDSS allergy shield cross-checks
- Drug-drug interaction risk analysis
- Schedule X narcotic PIN sign-off
- Manage chronic patient refill cycles

💳 Cashier (POS)

Terminal: Billing Counter

- 1-Click verified eRx import to cart
- Auto FEFO nearest-expiry batch pick
- Split payments (Cash, UPI QR, Card)
- Calculate cash change return
- Print 80mm GST tax receipt

📦 Inventory Manager

Terminal: Supply Hub

- Multi-batch FEFO stock ledger
- Record Goods Receipt Notes (GRN)
- 24/7 Cold-chain IoT fridge monitor
- Manage supplier directory & POs
- Track low-stock alerts (≤20 units)

🛵 Delivery Agent

Terminal: Logistics Bay

- Active dispatch Kanban route board
- Verify cold-bag insulated packaging
- 4-Digit customer OTP verification
- Record electronic Proof-of-Delivery
- Handover timestamp audit log

RUBRIC: MODULAR ARCHITECTURE

Modular Folder Architecture & Technical Baseline

📁 ENTERPRISE FOLDER STRUCTURE

```
Pharmacy_Management_System/
├── core/
│   ├── bridge.py       # Unified Python RPC Bridge (38 methods)
│   ├── db.py           # SQLite persistence layer (14 tables)
│   ├── api_client.py   # Presence heartbeat AppConnector
│   └── logger.py       # Tamper-evident audit logging engine
├── config/
│   └── settings.py     # Thread-safe Singleton dataclass
(config.json)
├── ui/
│   ├── desktop/        # Edge Chromium WebView2 60 FPS frontend
│   └── web/            # Full-featured FastAPI Web UI (1:1 parity)
├── apps/               # Standalone admin & operations GUI apps
├── tests/              # 24 Automated unit & integration tests
├── backups/            # Scheduled .db snapshots & CSV audit exports
├── main.py             # Hardware-accelerated desktop entrypoint
└── web_app.py          # Asynchronous FastAPI web application server
```

🏗️ 6-TIER DECOUPLED ARCHITECTURE

Tier 1: Presentation Layer
Modern HTML5, responsive CSS3 Glassmorphism, and vanilla ES6+ JavaScript. Clean SPA design without external framework bloat.

Tier 2: Desktop Runtime
Microsoft Edge Chromium (WebView2) via `pywebview 5.x` providing zero-latency 60 FPS hardware acceleration.

Tier 3: Web Application Server
Asynchronous `FastAPI 0.138.0` + `Uvicorn 0.49.0` ASGI backend with auto-generated Swagger API documentation.

Tier 4: Unified Bridge RPC
`PharmacyBridge` exposes 38 secure, thread-safe Python methods directly to frontend JavaScript.

Tier 5: Core Domain Engines
Autonomous engines: CDSS allergy cross-matcher, FEFO dynamic batch sorter, cold-chain IoT simulator, Daily Z-Report engine.

Tier 6: Persistence & Storage
SQLite 3 ACID relational database with 14 normalized tables, foreign keys, and timestamped automated backups.

RUBRIC: DATABASE / SYSTEM DESIGN

Database Design & System Integration (SQLite 3)

🗄️ CORE RELATIONAL TABLES (PART 1)

1. staff (PRIMARY KEY: id)
Credentials, assigned role, username, branch, and active status.

2. medicines (PRIMARY KEY: id)
SKU, generic name, category, Schedule (H1/X/OTC), rack, stock.

3. prescriptions (PRIMARY KEY: rx_id)
Patient name, doctor, items JSON, CDSS status, clinical sign-off.

4. invoices (PRIMARY KEY: invoice_id)
Customer, cashier, payment method, subtotal, GST, total amount.

5. delivery_orders (PRIMARY KEY: order_id)
Address, customer phone, rider, status, cold-bag check, OTP code.

6. patients (PRIMARY KEY: id)
Name, contact phone, chronic conditions, loyalty points, allergy records.

7. suppliers (PRIMARY KEY: id)
Distributor name, GSTIN, DL number, contact person, payment terms.

🗄️ COMPLIANCE & TELEMETRY TABLES (PART 2)

8. purchase_orders (PRIMARY KEY: po_id)
GRN shipment log, supplier, items JSON, total amount, inward status.

9. narcotic_register (PRIMARY KEY: id (AUTO))
Schedule X log: Doctor DMC Reg No, Pharmacist PIN, batch, quantity.

10. temperature_logs (PRIMARY KEY: id (AUTO))
24/7 cold-chain telemetry: Sensor ID, temperature (2-8°C), alert flag.

11. refill_schedules (PRIMARY KEY: id)
Chronic adherence: Patient, medicine, last dispensed, next due date.

12. active_sessions (PRIMARY KEY: session_id)
Live presence: User ID, role, terminal name, last heartbeat timestamp.

13. audit_logs (PRIMARY KEY: id (AUTO))
Immutable security log: User, role, action, details, severity level.

14. z_reports (PRIMARY KEY: z_report_id)
Daily closing: Opening float, cash sales, payouts, variance, signed status.

RUBRIC: CRUD OPERATIONS & FUNCTIONALITY

Comprehensive CRUD Operations Implementation

COMPLETE CRUD IMPLEMENTATION ACROSS ALL CORE MODULES

Module / Entity	CREATE	READ	UPDATE	DELETE
Staff & Users	Add staff user modal (ID, Name, Role, Branch)	Staff credentials list & live online presence	Edit staff role, password, branch & status	Delete unauthorized staff user with audit log
Medicines Catalog	Add medicine modal (SKU, Category, Batch)	Catalog table with FEFO visual countdown badges	Adjust physical stock delta (+/- units modal)	Delete expired/discontinued medicine SKU
Clinical Prescriptions	Inward e-Prescription modal with medicine items	Prescriptions review queue with CDSS allergy scan	Update verification status (Approve / Reject)	Archive processed prescriptions into sales ledger
Narcotic Register	Add Schedule X dispensation with Doctor Reg No	Read narcotic audit ledger with search filter	Update verified status with 4-digit Pharmacist PIN	Restricted: Immutable audit records (CDSCO law)
Invoices & POS	Generate sales invoice with 5% CGST/SGST split	Sales ledger history with 1-click receipt modal	Update patient loyalty points upon purchase	Void invoice requires supervisor override
Deliveries & Logistics	Create home delivery order with customer phone	Dispatch Kanban board (Packaging/Out/Delivered)	Advance delivery status & verify 4-digit OTP	Cancel unassigned delivery dispatch task
Daily Z-Report Close	Generate official Z-Report ID (Z-YYYYMMDD-XXXX)	Read gross sales, cash total, digital collections	Update counted cash & calculate live variance	Permanent archive: Immutable signed registry

RUBRIC: MODULES & QUALITY ALIGNMENT

Modules Aligned with TQM Quality Goals

🩺 Clinical Decision Support (CDSS)

ALIGNED TQM GOAL: ZERO MEDICATION DISPENSING DEFECTS

- Real-time allergen cross-checks (Penicillin, NSAIDs, Sulfa).
- Molecule-molecule drug interaction checker.
- Double sign-off for Schedule X narcotics (Pharmacist PIN + Doctor Reg).
- Patient chronic disease and allergy EHR integration.

📦 FEFO Inventory & Cold-Chain IoT

ALIGNED TQM GOAL: ZERO BIOLOGICAL POTENCY LOSS & ZERO DEAD-STOCK

- Strict First-Expiry, First-Out batch sequencing algorithms.
- Visual countdown tags (<30d Critical, <90d Attention).
- 24/7 continuous IoT refrigerator telemetry (2.0°C–8.0°C).
- Goods Receipt Notes (GRN) dockside supplier verification.

💳 Cashier Express POS & GST Billing

ALIGNED TQM GOAL: SERVICE SPEED, TAX INTEGRITY & ZERO CASH LEAKAGE

- 1-Click verified eRx import from clinical queue.
- Multi-tender checkout: Cash change calculator, Dynamic UPI QR, Card.
- Automated statutory 5% GST calculation (2.5% CGST + 2.5% SGST).
- Instant 80mm thermal receipt printing and sales ledger.

🛵 Delivery Dispatch Fleet (e-POD)

ALIGNED TQM GOAL: 100% LAST-MILE HANDOVER FIDELITY

- Visual 3-column route Kanban board (Packaging, Out, Delivered).
- Cold-bag insulated packaging verification checklist.
- 4-Digit customer SMS OTP verification before order handoff.
- Automatic GPS arrival and handover timestamp logging.

RUBRIC: INITIAL UI & MAIN WORKFLOW

Initial UI & End-to-End System Workflow

STEP 1

Stage 1: Prescription Intake

- Doctor e-Prescription uploaded
- Patient EHR allergies scanned
- Enters pending review queue

STEP 2

Stage 2: Clinical CDSS Gate

- Pharmacist runs CDSS scan
- Allergy & interaction checks
- Schedule X PIN authenticated
- Status updated to 'Approved'

STEP 3

Stage 3: POS Express Billing

- 1-Click import to Cashier cart
- Auto FEFO batch picking
- Dynamic UPI QR / Cash change
- Thermal 80mm GST receipt printed

STEP 4

Stage 4: Cold-Chain & Fleet

- Cold-bag checklist verified
- Dispatched on Kanban route
- Customer gives 4-digit OTP
- Marked 'Delivered' (e-POD)

STEP 5

Stage 5: Daily Z-Report Close

- Cash drawer counted
- Variance calculated (₹0.00)
- Shift signed & closed in SQLite
- CSV exported for accounting

🎨 USER INTERFACE DESIGN & DUAL DEPLOYMENT PARITY

- Desktop UI: Rendered via Microsoft Edge Chromium WebView2 engine with hardware acceleration (60 FPS) and zero native bridge latency.
- Web UI: Hosted asynchronously via FastAPI (Uvicorn), providing 100% feature parity on Chrome, Safari, Edge, tablets, and mobile.
- Design System: Custom dark-mode Glassmorphism with CSS variables, responsive side navigation, interactive modals, and live clock tickers.

FAIL-SAFE QUALITY GATES

Poka-Yoke: Mistake-Proofing Healthcare Operations

1. CDSS Patient Allergy Shield

POKA-YOKE MECHANISM: AUTOMATED BLOCKING GATE

The cashier or pharmacist cannot bypass or dispense drugs containing penicillin, NSAIDs, or sulfas if the patient EHR contains registered hypersensitivity. Prevents fatal anaphylaxis before billing.

2. Molecule Interaction Guard

POKA-YOKE MECHANISM: CROSS-REFERENCE ALERT

When combining multiple drugs in a prescription (e.g. Warfarin + Augmentin, Glycomet + Alcohol), the Clinical Decision Support System triggers a severe warning modal preventing toxic dosing.

3. Schedule X Narcotics Dual PIN

POKA-YOKE MECHANISM: MANDATORY DUAL AUTHENTICATION

Narcotic sedatives (e.g. Alprax, Diazepam) cannot be added to the POS cart without doctor registration number validation and a 4-digit pharmacist authorization PIN, eliminating illegal diversion.

4. 4-Digit OTP Delivery Handshake

POKA-YOKE MECHANISM: CRYPTOGRAPHIC HANDSHAKE

A delivery agent cannot mark an order as 'Delivered' simply by clicking a button. The customer must provide a 4-digit SMS OTP, guaranteeing medicines are handed to the rightful patient.

RUBRIC: WORKING APPLICATION / DEMO

Working Application Live Demonstration Guide

Step 1: Application Launch

Desktop Launch:

`python main.py`

(Hardware-accelerated Edge Chromium WebView2 window)

Web Server Launch:

`python web_app.py`

(Access at http://127.0.0.1:8000)

Swagger API Documentation:

Access at http://127.0.0.1:8000/docs

Step 2: Role Walkthrough

1. Login as Admin (`admin`/`admin123`):

   Inspect RBAC, terminals, audit log.

2. Switch to Pharmacist:

   Review eRx, trigger CDSS allergy shield.

3. Switch to Cashier:

   1-Click Rx import, FEFO batch selection, UPI QR modal, print 80mm GST receipt.

Step 3: Quality Gates & Close

4. Switch to Inventory Manager:

   Demonstrate 2–8°C thermal spike alarm, record Goods Receipt Note (GRN).

5. Switch to Delivery Rider:

   Advance Kanban, verify 4-digit OTP.

6. Return to Admin / Cashier:

   Perform Daily Z-Report cash reconciliation, balance till, print official Z-slip.

RUBRIC: IMPLEMENTED FUNCTIONALITY

Quality Verification & 24 Automated Test Results

24 / 24

Automated Tests Passed

100% pass rate across end-to-end unit, integration, and security test suites.

0%

Medication Dispensing Errors

Achieved via automated Poka-Yoke CDSS allergy and interaction barriers.

100%

Cold-Chain Integrity (2–8°C)

Continuous 30s background telemetry streaming with instant excursion alerts.

₹ 0.00

Cash Drawer Variance

Achieved on balanced registers via Daily Z-Report mathematical reconciliation.

📊 TRADITIONAL PHARMACY VS. AEGISPHARM TQM BENCHMARK

Quality Dimension	Traditional Retail Pharmacy (Manual)	AegisPharm Enterprise (TQM Driven)
Prescription Safety	Manual pharmacist check; vulnerable to handwriting errors	Automated CDSS cross-checks with patient allergy EHR
Inventory Picking	Random or FIFO; older batches expire on shelves	Algorithmic FEFO (First-Expiry, First-Out) guarantee
Cold-Chain Monitoring	Infrequent manual thermometer checks; silent spoilage	Continuous 30s IoT telemetry with automatic alarm alerts
Shift Cash Balancing	Paper ledger estimations; undetected shrinkage	Daily Z-Report mathematical reconciliation & signed Z-slip

REVIEW 1 SUMMARY & STRATEGIC CONCLUSION

Review 1 Milestones Fully Delivered

✓ Problem Statement & Quality Objectives: Clearly formulated and aligned with TQM principles.

✓ Formal SRS Document: Comprehensive IEEE 830 functional and non-functional requirements specified.

✓ Scope & User Roles: 5 specialized RBAC workstations with granular access control implemented.

✓ Modular Folder Architecture: 6-tier decoupled structure separating core logic, config, and UIs.

✓ Database & CRUD Functionality: 14 relational tables in SQLite with complete CRUD operations working live.

✓ Working Application & Demo: Operational desktop (Edge Chromium) and web (FastAPI) applications demonstrated.

🙏 **THANK YOU RESPECED FACULTY & EVALUATORS**

Sincere gratitude to the Computer Science & Management faculty at Amrapali University for their academic guidance.

💬 **The floor is now open for questions, code inspection, and live application demonstration. (Piyush Bhatt)**